

Cambridge (CIE) IGCSE Additional Maths



Your notes

Equations, Inequalities & Graphs

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- * [Modulus Functions](#)
- * [Graphs of Cubic Polynomials](#)



Your notes

Modulus Functions

Sketching Modulus Graphs

What is a modulus function?

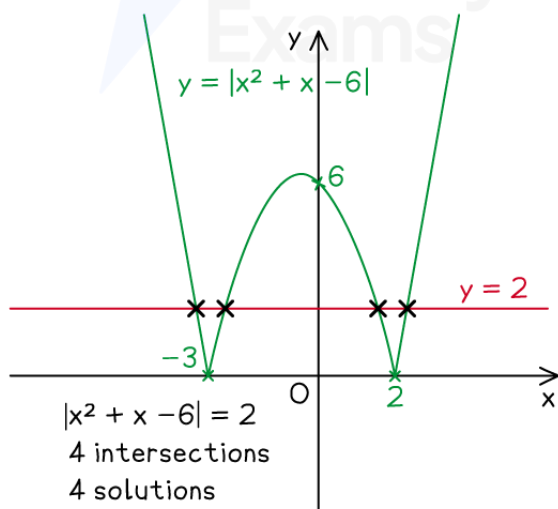
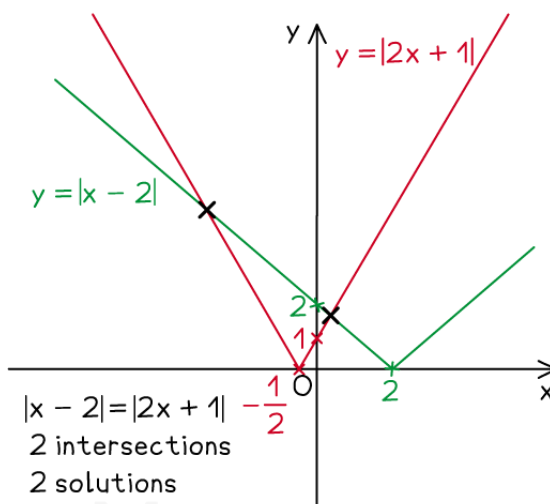
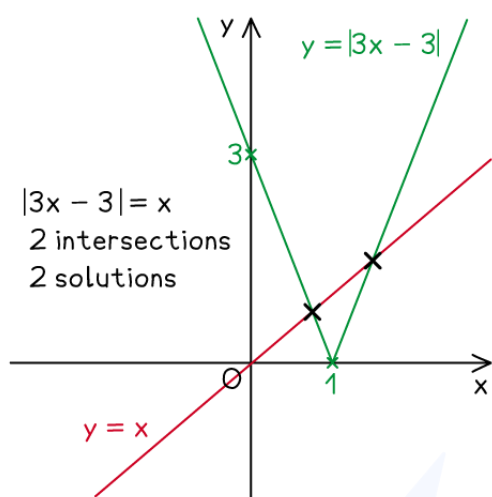
- The **modulus function** makes any 'input' **positive**
 - This is sometimes called the **absolute value** (of the input)
 - The modulus function is indicated by a pair of vertical lines being written around the input
 - Similar to how brackets are used
 - e.g. $|7| = 7$, $|-7| = 7$

What types of modulus graphs will I need to sketch?

- Modulus graphs required will be of **linear**, or **quadratic** form
 - **Linear** form will be $|ax + b|$
 - **Quadratic** form will be $|ax^2 + bx + c|$
- Often, there will be two graphs to sketch as this helps with **solving equations** involving modulus functions
 - In linear form, equations could be of the form $|ax + b| = |cx + d|$
 - One side of the equation may not involve the modulus
 - One side may have a constant term only (i.e. $b = 0$ and/or $d = 0$)
 - In quadratic form, equations could be of the form $|ax^2 + bx + c| = d$
- In both cases, graphs of the left hand side and right hand side drawn on the same diagram will reveal the number of intersections of the graphs (and so the number of solutions to the equation)



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How do I sketch the graph of the modulus of a function: $y = |f(x)|$?

STEP 1

Pencil in the graph of $y = f(x)$

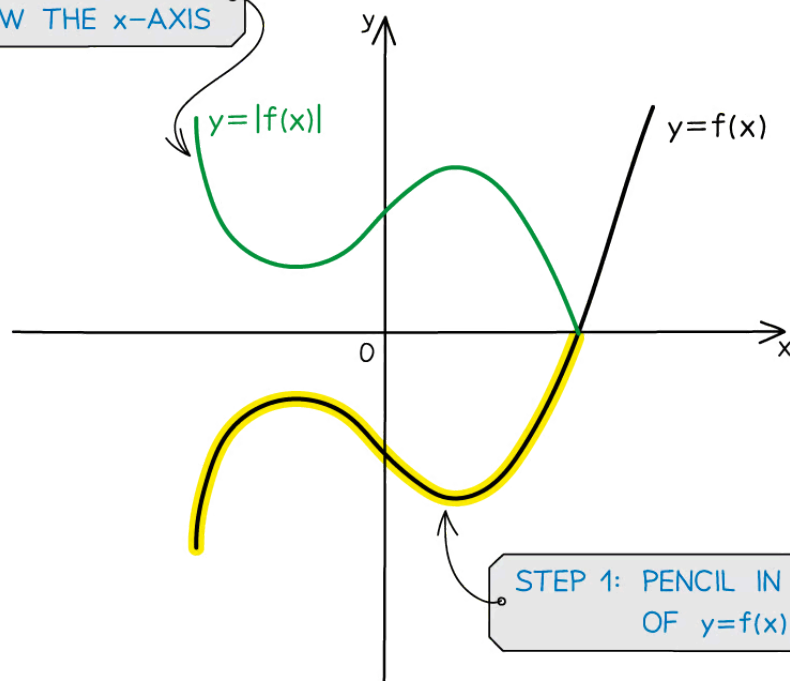
STEP 2

Reflect anything **below** the **x**-axis, in the **x**-axis, to get $y = |f(x)|$



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STEP 2: REFLECT ANYTHING
BELOW THE x -AXIS



STEP 1: PENCIL IN GRAPH
OF $y = f(x)$

- Note in particular that the y -axis intercept, if negative on the graph of $y = f(x)$ will be positive on the graph of $y = |f(x)|$
- At the x -axis intercepts, the graph will have a sharp ∇ -shape
 - this is **not** a smooth curve like with a turning point
 - if the graph is of the form $y = -|f(x)|$ then the graph would be wholly negative and the x -axis intercepts would have a sharp $\wedge$ -shape



Worked Example



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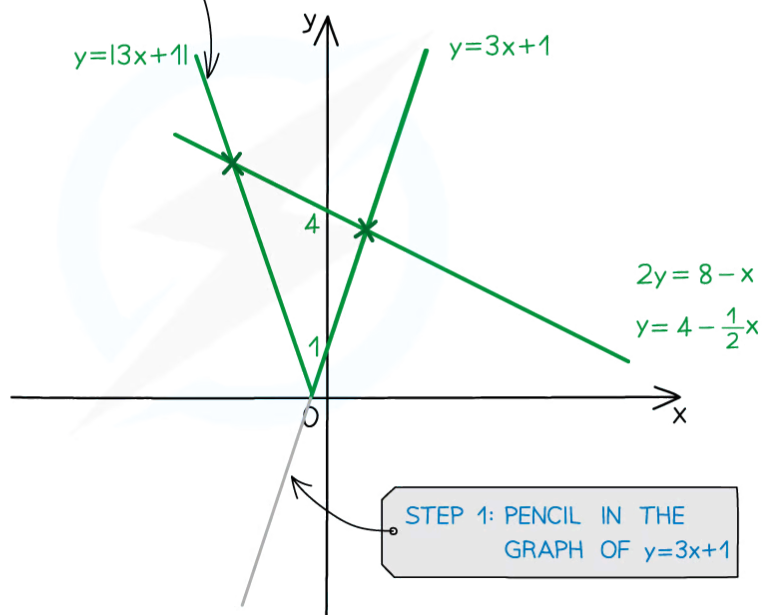


(a) On the same diagram sketch the graphs of $y = |3x + 1|$
and $2y = 8 - x$

(b) Hence write down how many solutions there are to the
equation $4 - \frac{1}{2}x = |3x + 1|$

(a)

STEP 2: REFLECT NEGATIVE
PART IN THE x-AXIS



STEP 1: PENCIL IN THE
GRAPH OF $y = 3x + 1$

(b) 2

THE GRAPHS CROSS TWICE

$$4 - \frac{1}{2}x = -(3x + 1)$$

$$4 - \frac{1}{2}x = 3x + 1$$

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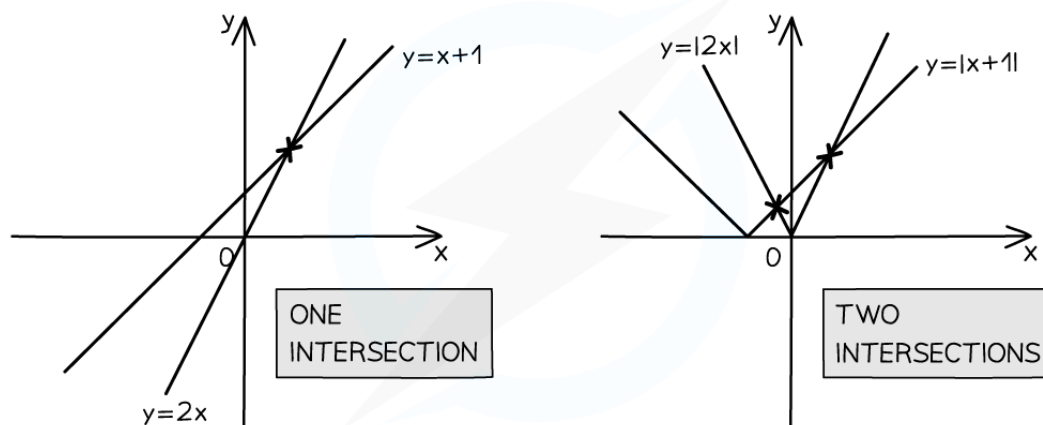


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Solving Modulus Equations

Why are graphs needed to solve modulus equations?

- **Sketching** the graphs of two modulus function(s) on the same diagram quickly reveals
 - the number of solutions there are to the equation
 - which parts of the graph (equation) - either the 'normal' part or the 'reflected' part - will be needed to solve equations
- For example, two non-parallel linear graphs would intersect
 - the solution to the linear functions being equal to each other would have one solution
 - if a **modulus** is involved there **could** be **more** than one intersection/solution



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How do I solve modulus equations?

STEP 1 **Sketch** the graphs including any **modulus** (reflected) parts

STEP 2 **Locate** the graph **intersections**

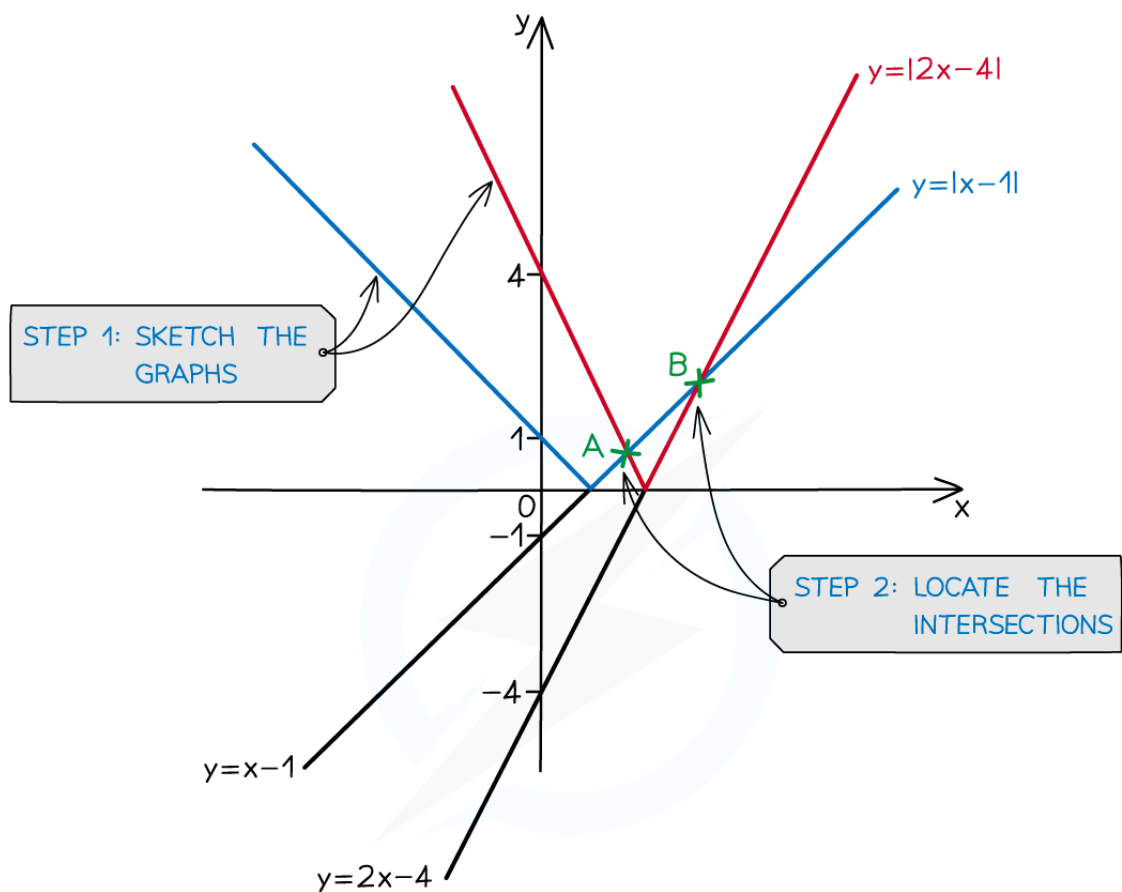
STEP 3

Determine which part of each graph ('normal' or 'reflected' part) is needed to solve the equation **Solve** the appropriate equation(s)



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e.g. SOLVE $|2x-4| = |x-1|$



STEP 3: SOLVE THE APPROPRIATE EQUATIONS

A: $x-1 = -(2x-4)$
 $3x = 5$
 $x = \frac{5}{3}$

• "NORMAL" PART OF $x-1$
 • "REFLECTED" PART OF $2x-4$

B: $x-1 = 2x-4$
 $x = 3$

"NORMAL" PART FOR BOTH

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How do I solve modulus inequalities?

- The process is very similar to that as solving equations - with the graph sketching being essential

STEP 1 **Sketch** the graph(s) including any **modulus** (reflected) parts

STEP 2 **Locate** the graph **intersections** (or **X**-axis intercepts if zero on one side)

STEP 3 Determine which part(s) of the graph(s) satisfy the inequality (highlight any on the graph) Find the **intersections** (by solving equation(s))

STEP 4 Write the final answer(s) down, being careful with the use of $<$, $>$, $\leq$ and/or $\geq$

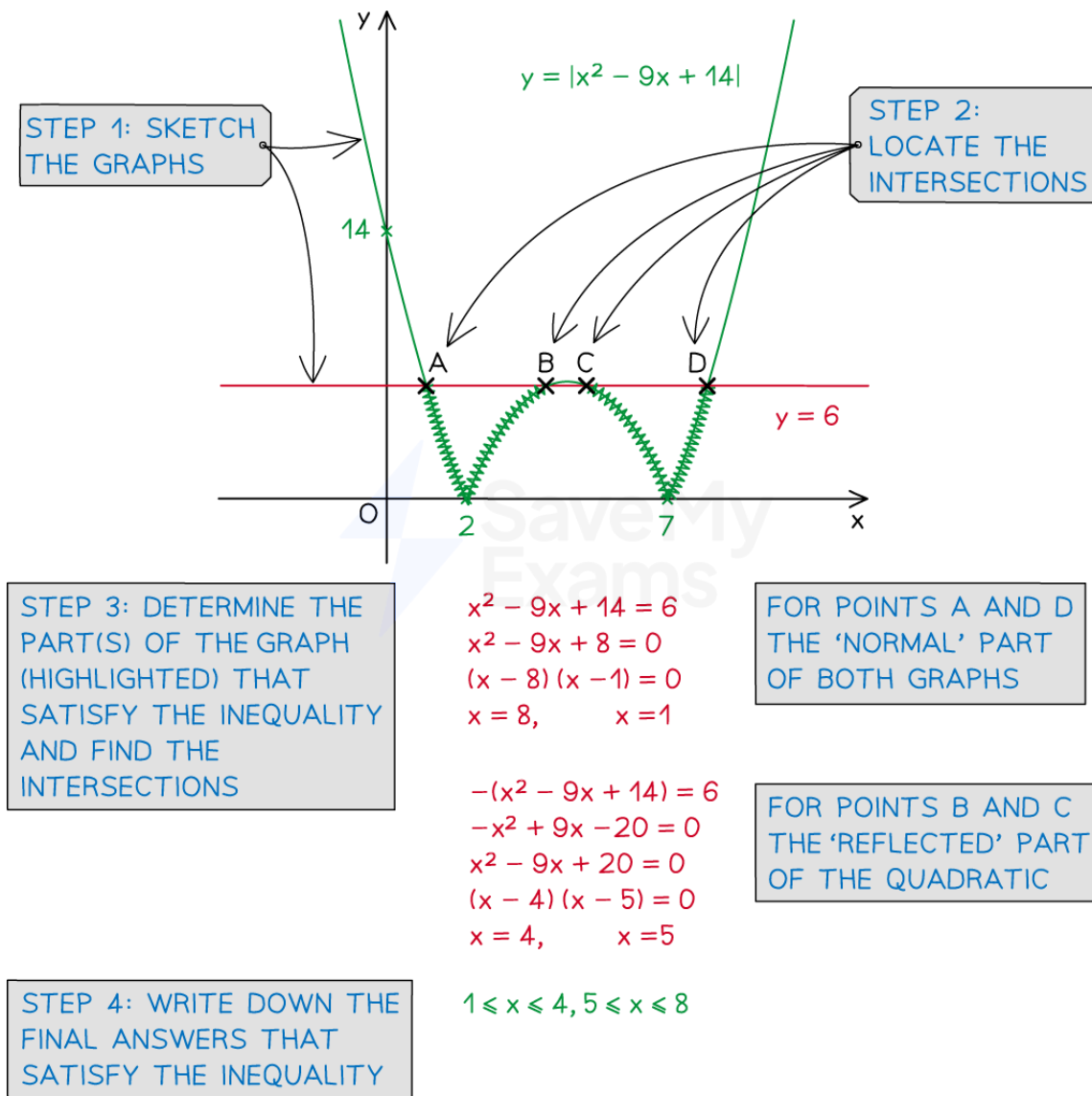


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e.g. SOLVE $|x^2 - 9x + 14| \leq 6$



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Examiner Tips and Tricks

- Sketching the graphs is important as solving algebraically can lead to invalid solutions



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- For example, $x = 1$ is a solution to $x - 4 = 2x - 5$
- but $x = 1$ is not a solution to $|x - 4| = 2x - 5$
 - (substitute $x = 1$ into both sides and see why it does not work)



Worked Example

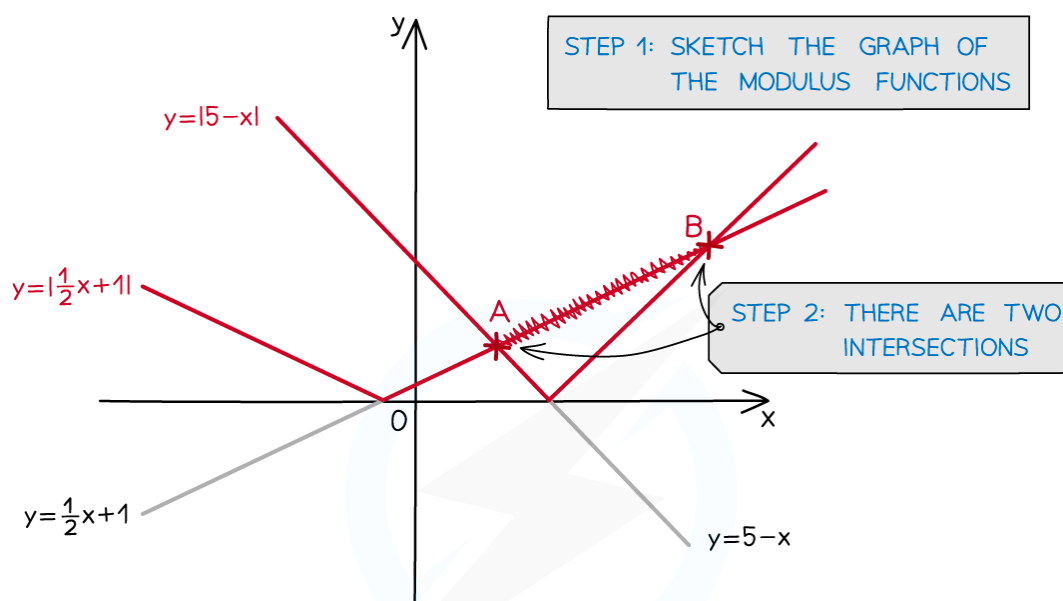


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? (a) Solve the equation $\left| \frac{1}{2}x + 1 \right| = |5 - x|$

(b) Hence solve the inequality $\left| \frac{1}{2}x + 1 \right| \geq |5 - x|$

(a)



A: $\frac{1}{2}x + 1 = 5 - x$
 $\frac{3}{2}x = 4$
 $x = \frac{8}{3}$

STEP 3: POINT A IS THE NORMAL PART OF BOTH EQUATIONS

B: $\frac{1}{2}x + 1 = -(5 - x)$
 $6 = \frac{1}{2}x$
 $x = 12$

STEP 3: POINT B IS THE NORMAL PART OF $\frac{1}{2}x + 1$ BUT THE (REFLECTED) NEGATIVE PART OF $5 - x$

∴ SOLUTIONS TO $\left| \frac{1}{2}x + 1 \right| = |5 - x|$ ARE $x = \frac{8}{3}$ AND $x = 12$

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(b) $\frac{8}{3} \leq x \leq 12$

SHADE ON YOUR DIAGRAM THE PART OF THE GRAPH
WHERE $|\frac{1}{2}x + 1| \geq 15 - x$
THIS SHADING IS BETWEEN THE TWO INTERSECTIONS
WHICH YOU FOUND IN PART a)

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Graphs of Cubic Polynomials

Graphs of Cubic Polynomials

What is a cubic polynomial?

- A **cubic polynomial** is a function of the form $ax^3 + bx^2 + cx + d$
 - a , b , c and d are constants
 - it is a polynomial of degree 3
 - so b , c and/or d could be zero
- To sketch the graph of a cubic polynomial it will need to be in factorised form
 - e.g. $(2x - 1)(x + 2)(x - 3)$ is the factorised form of $2x^3 - 3x^2 - 11x + 6$

What does the graph of a cubic polynomial look like?

- In general the graph of a cubic polynomial will take one of the four forms
 - All are **smooth** curves that take some practice to sketch!



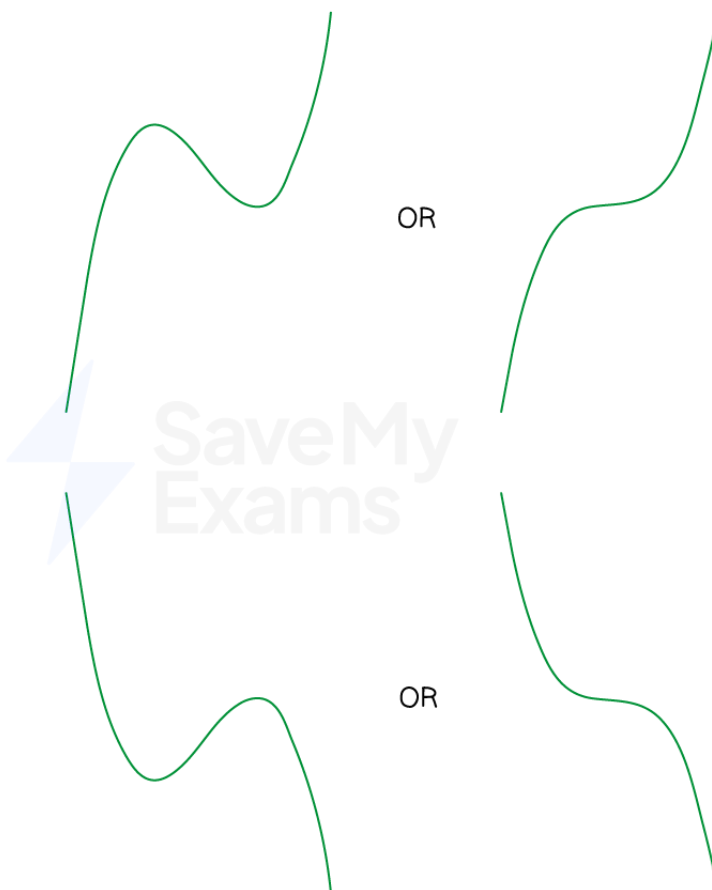
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POSITIVE CUBIC
POLYNOMIALS

$$a > 0$$

NEGATIVE CUBIC
POLYNOMIALS

$$a < 0$$

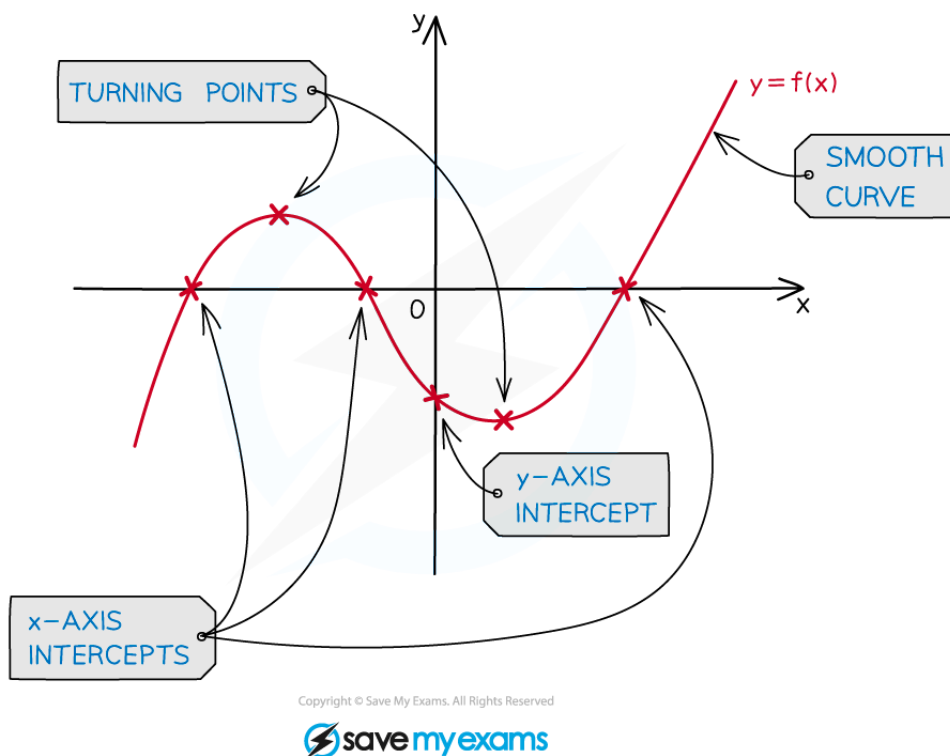


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- The exact form a particular cubic polynomial will depend on
 - The number (and value) of **roots** (x -axis intercepts) of the cubic polynomial
 - The y -axis intercept
 - The sign of the coefficient of the x^3 term (a)
 - If $a > 0$ the graph is a **positive cubic** ('starts' in the third quadrant, 'ends' in the first)
 - If $a < 0$ the graph is a **negative cubic** ('starts' in the second quadrant, 'ends' in the fourth)
 - **Turning points**



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How do I sketch the graph of a cubic polynomial?

STEP 1 Find the y -axis intercept by setting $x = 0$

STEP 2 Find the x -axis intercepts (roots) by setting $y = 0$ (Any repeated roots will mean the graph *touches* - rather than *crosses* - the x -axis)

STEP 3 Consider the shape of the graph - is it a **positive cubic** or a **negative cubic**? Where does the graph 'start' and 'end'?

STEP 4 Consider where any turning points should go

STEP 5 Sketch the graph with a *smooth* curve, labelling points where the graph intercepts the x and y axes



Examiner Tips and Tricks

- In the exam, a cubic polynomial that requires sketching will be given in factorised form



Your notes

- i.e. as the product of three linear factors
- a factor could be repeated
- Build your sketch up as you work through each step to gradually build a mental picture of it
- If you need to redraw the graph for your final answer, that's fine!



Worked Example

Sketch the graph of $y = (2x - 1)(x - 3)^2$.

STEP 1 - y -axis intercept

$$y = (-1)(-3)^2 = -9$$

STEP 2 - x -axis intercepts

$$2x - 1 = 0, x = \frac{1}{2}$$

$$x - 3 = 0, x = 3 \text{ (repeated)}$$

STEP 3 - shape, 'start'/'end'

$a > 0$ ($a = 2$) so it is a positive cubic $x = 3$ is a repeated root so the graph will touch the x -axis at this point

STEP 4 - turning points

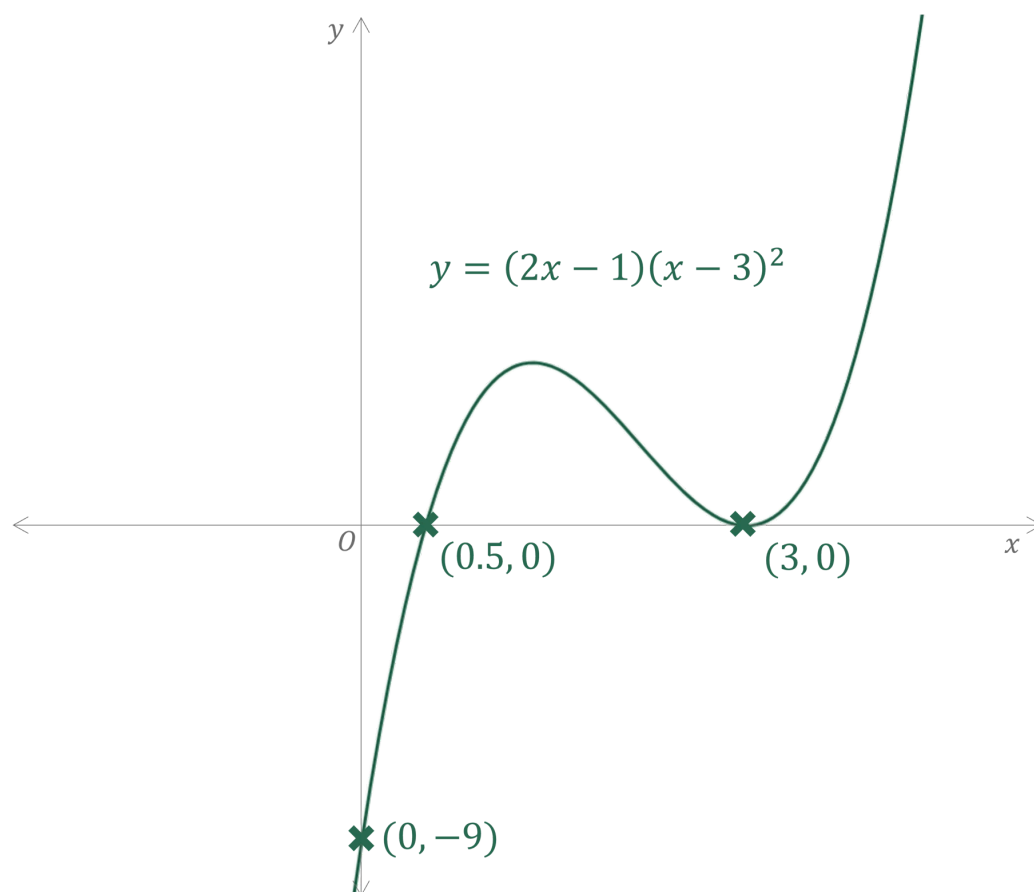
One turning point (minimum) will need to be where the curve touches the x -axis The other

(maximum) will need to be between $x = \frac{1}{2}$ and $x = 3$

STEP 5 - smooth curve with labelled intercepts



Your notes



Modulus Cubic Graphs

What is a modulus cubic graph?

- A (factorised) **cubic** polynomial is of the form $f(x) = a(x - b)(x - c)(x - d)$
- The graph of $y = f(x)$ must cross the x -axis at least once
 - therefore y must take both positive and negative values
- The **modulus** cubic graph, $y = |f(x)|$ will mean all values of y are positive
 - Any negative values become their positive equivalents
 - e.g. If $f(x) = x^3$ then $f(-1) = (-1)^3 = -1$, but $|f(-1)| = |(-1)^3| = |-1| = 1$
- A modulus cubic graph will not have any negative values

- the graph will not cross the X -axis
- the graph will touch the X -axis (at least once)

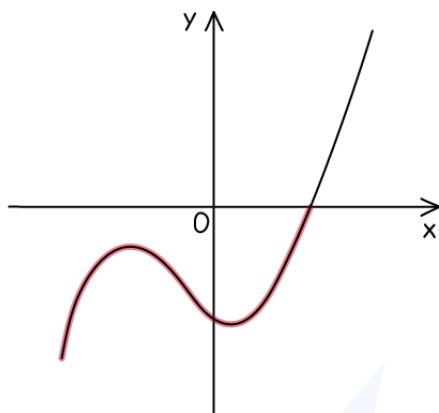


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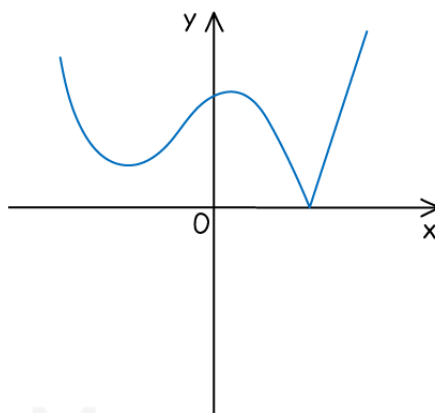
How do I sketch a modulus cubic graph?

- Sketch** the graph of the (original) cubic polynomial, $y = f(x)$
- Any parts of this graph that are **below** the X -axis should be **reflected** in the X -axis to sketch the graph of $y = |f(x)|$

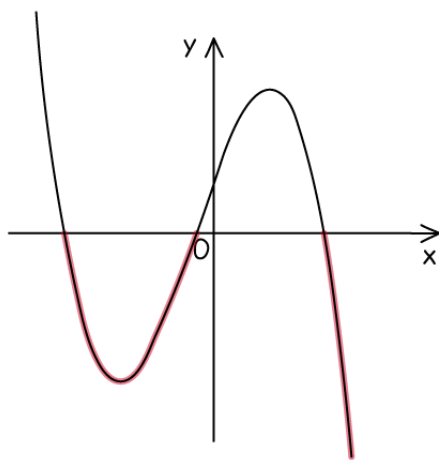
$$y = f(x)$$



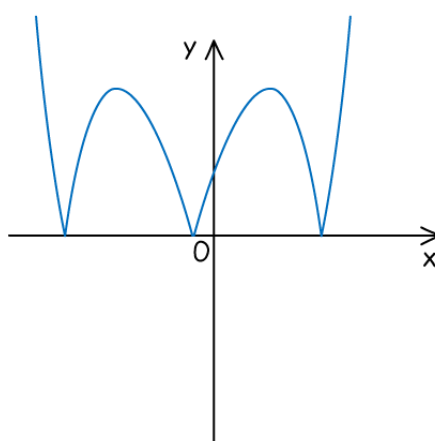
$$y = |f(x)|$$



$$y = g(x)$$



$$y = |g(x)|$$



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- The points at which a modulus graph touches the X -axis are the same as the points at which the original graph intercepts the X -axis (i.e. the roots of $f(x)$)
 - **Label** these points, and the y -axis intercept, on a sketch
- Notice that the points at which a modulus graph touches the X -axis are **not** smooth
 - they are 'pointy' (V-shaped)
 - turning points are smooth (U-shaped)

How do I find a cubic function from its modulus graph?

- To deduce a cubic expression from its modulus graph consider
 - whether the (original) expression could be a **positive** or **negative** cubic
 - a positive cubic 'starts' in the third quadrant and 'ends' in the first
 - a negative cubic 'starts' in the second quadrant and 'ends' in the fourth
 - a negative cubic can have a "-" at the start of its expression
 - the X -axis intercepts - the roots
 - for the roots b , c and d , write the factors $(x - b)(x - c)(x - d)$
 - the y -axis intercept - to deduce the expression in the form $a(x - b)(x - c)(x - d)$
 - lots of cubic functions have the same roots but have different coefficients
 - the y -axis intercept should be the product $a \times -b \times -c \times -d = -abcd$
 - a may often, but not always, be 1



Examiner Tips and Tricks

- Sketching the (original) graph of $y = f(x)$ first is often helpful rather than trying to sketch the modulus graph from scratch
- If asked to find a cubic function from a given graph
 - there may be more than one possibility
 - consider which parts of the graph may have been reflected in the X -axis
 - could it be either a positive or negative cubic polynomial?

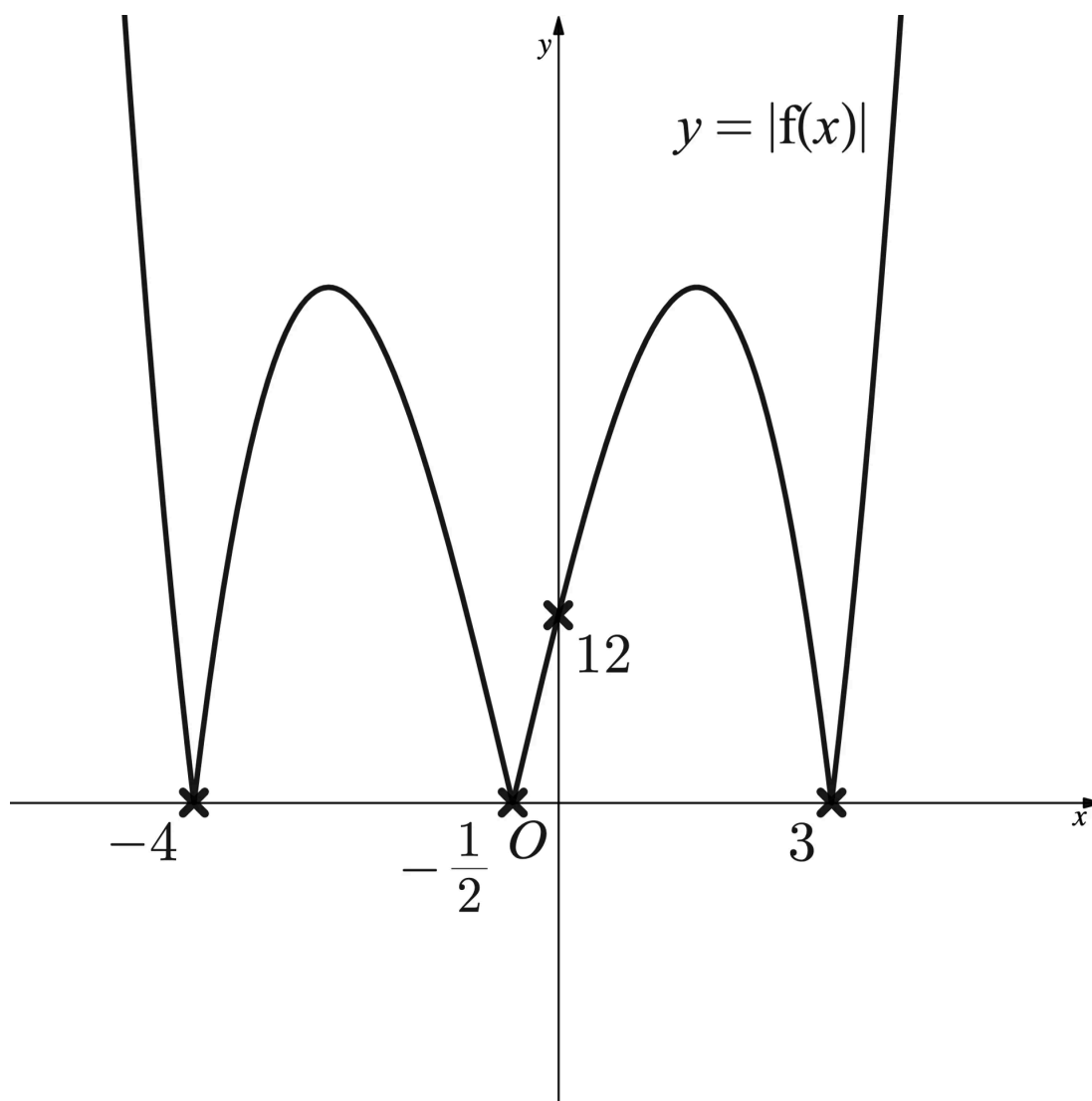


Your notes



Worked Example

The diagram below shows the graph of $y = |f(x)|$ where $f(x)$ is a **positive** cubic polynomial.

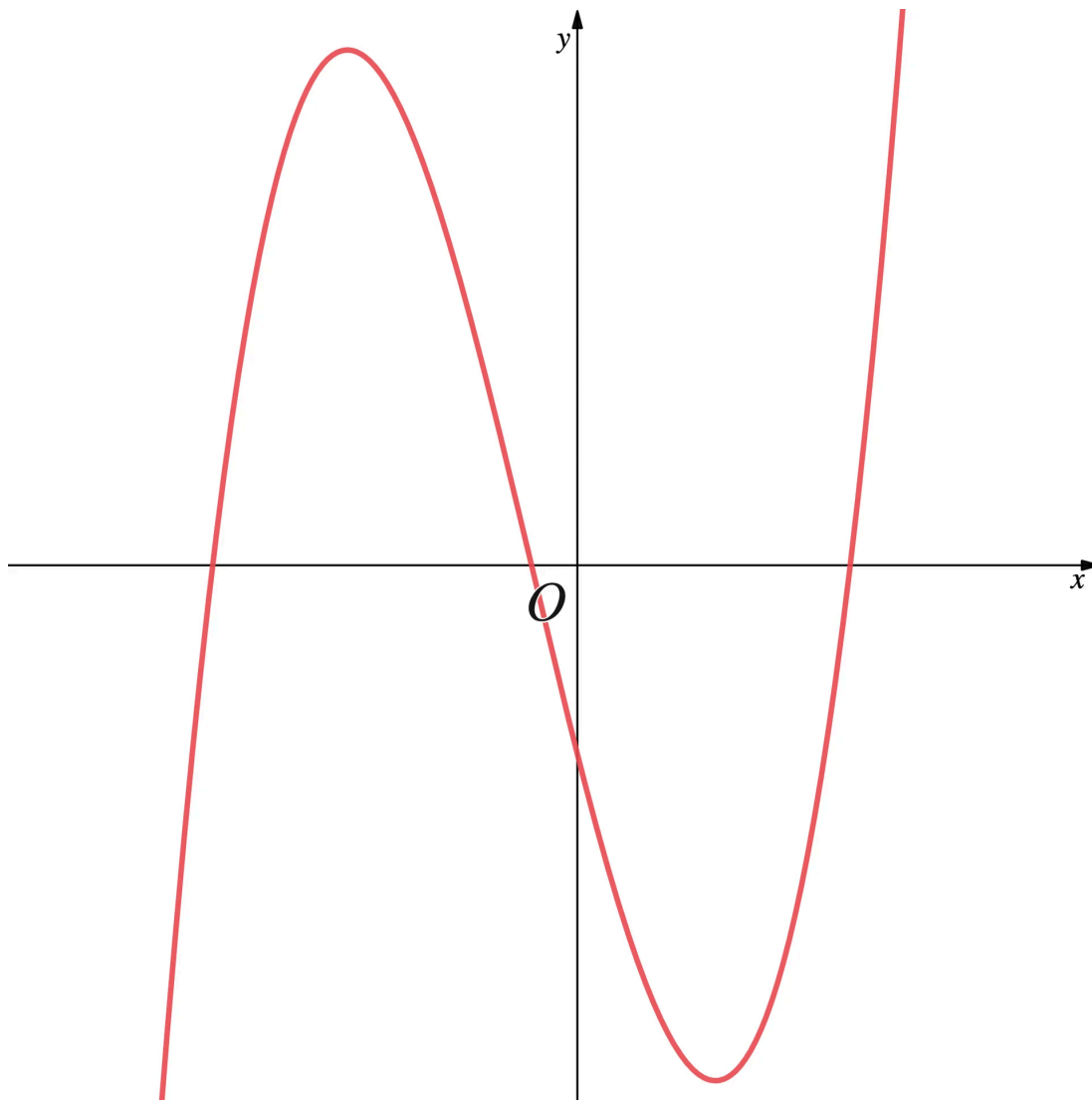


Deduce an expression for $f(x)$ leaving your answer in factorised form, with each factor having integer coefficients.



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$f(x)$ is a positive cubic polynomial so the graph should 'start' in the third quadrant and 'end' in the first. A quick sketch (without labels) can help.



The x -axis intercepts indicate the roots.

$$(x + 4)\left(x + \frac{1}{2}\right)(x - 3)$$



Your notes

The expression for $f(x)$ is of the form $a(x+4)\left(x+\frac{1}{2}\right)(x-3)$. Use the y -axis intercept to deduce the value of a . From the sketch of $y=f(x)$ the y -axis intercept is -12 (not 12).

$$\begin{aligned} a \times 4 \times \frac{1}{2} \times -3 &= -12 \\ -6a &= -12 \\ a &= 2 \end{aligned}$$

The final answer requires each factor to have integer coefficients so multiply 2 into $\left(x+\frac{1}{2}\right)$ and write down $f(x)$.

$$f(x) = (x+4)(2x+1)(x-3)$$

Solving Cubic Inequalities Graphically

What is a cubic inequality?

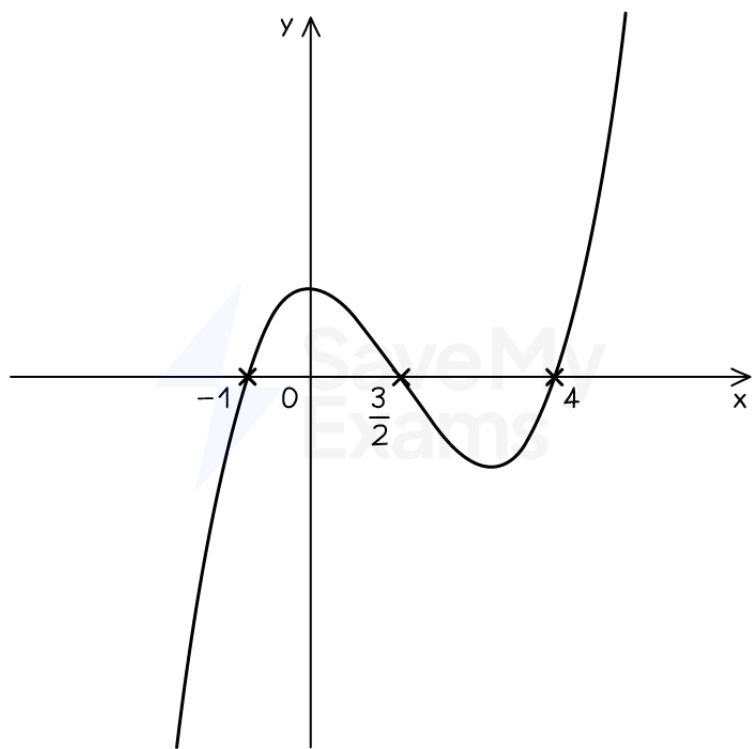
- A cubic function is of the form $f(x) = ax^3 + bx^2 + cx + d$ where a , b , c and d are constants
- A **cubic inequality** can be any of the following
 - $f(x) > 0$
 - $f(x) \geq 0$
 - $f(x) < 0$
 - $f(x) \leq 0$
- An inequality may need **rearranging** into one of these forms first before solving
 - Furthermore, solving cubic equations graphically is easiest when the expression has been **factorised**
 - e.g. for a cubic with three (real) roots this would be $(x-p)(x-q)(x-r)$ where p , q and r are the roots

How do I solve a cubic inequality graphically?

STEP 1 If need be, rearrange the inequality so that one side of the inequality is zero This should leave a **cubic polynomial** on the other side **Factorise** the cubic polynomial if required e.g.

$$(2x - 3)(x - 4)(x + 1) \leq 0$$

STEP 2 **Sketch** the graph of the cubic polynomial The **X**-axis intercepts (roots) are crucial to finding the solution but the **y**-axis intercept is not needed e.g.



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STEP 3

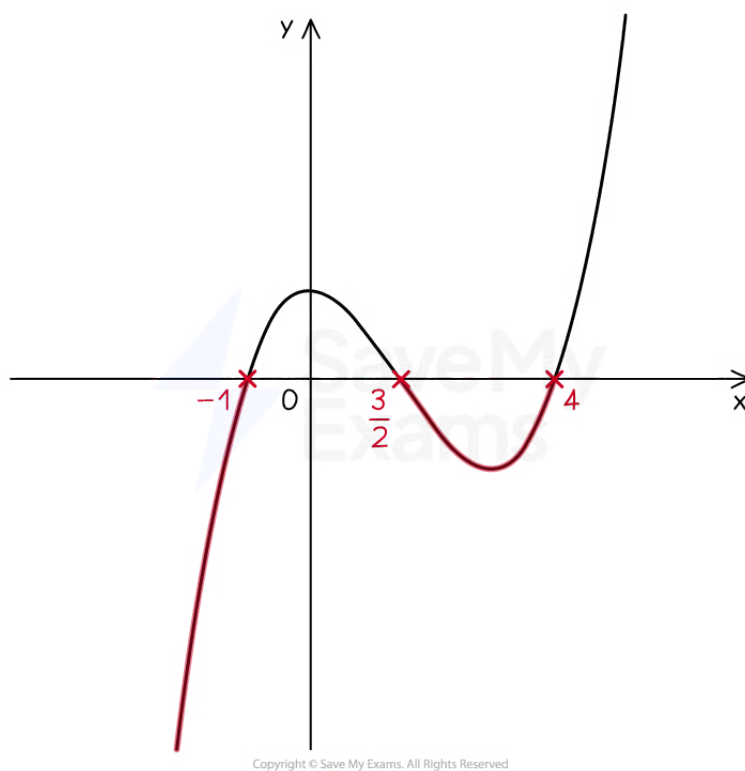
Identify the part(s) of the graph that **satisfy** the inequality

Highlighting this on the sketch will help

Consider whether you need to **include** ($\leq$, $\geq$) or **exclude** ($<$, $>$) the **roots** e.g.



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STEP 4

Write down the **solutions** to the inequality

e.g. $x \leq -1, \frac{3}{2} \leq x \leq 4$



Worked Example

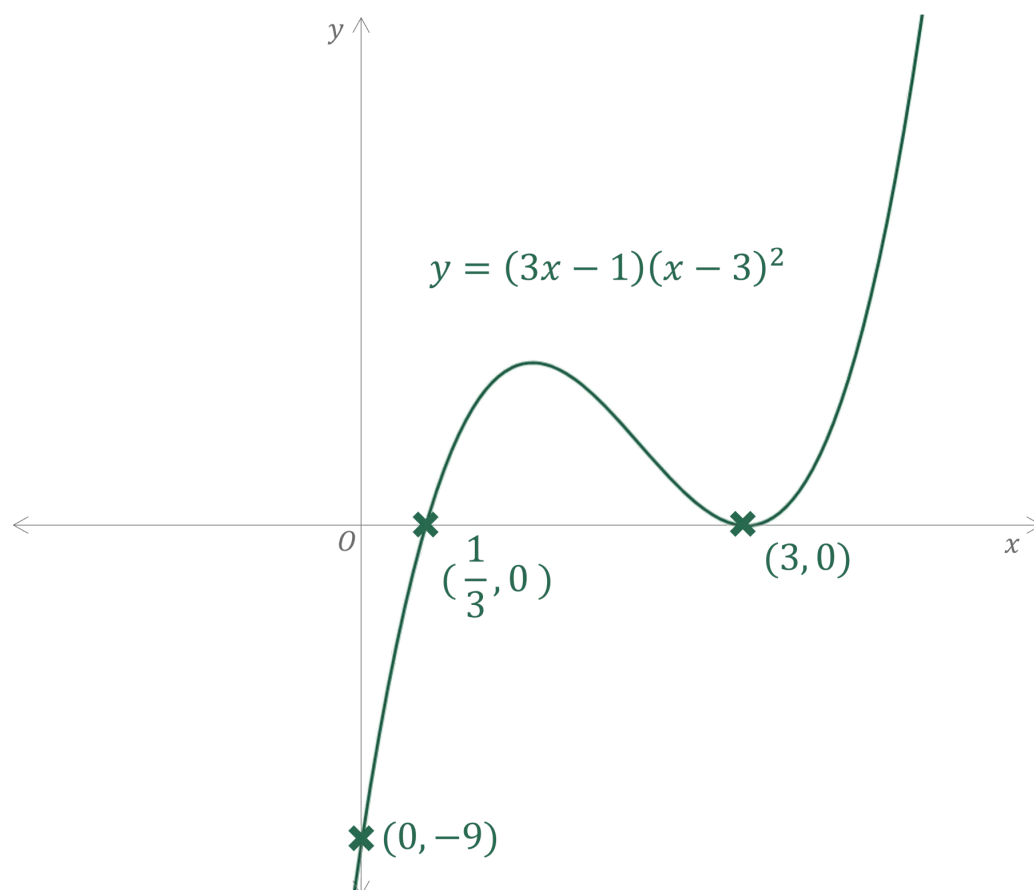
a) Sketch the graph of $y = f(x)$ where $f(x) = (3x - 1)(x - 3)^2$.

This is a **positive** cubic polynomial with roots $x = \frac{1}{3}$ and $x = 3$

$x = 3$ is a **repeated** root so the graph will **touch** the x -axis at this point



Your notes



a) Hence solve the inequality $f(x) > 0$.

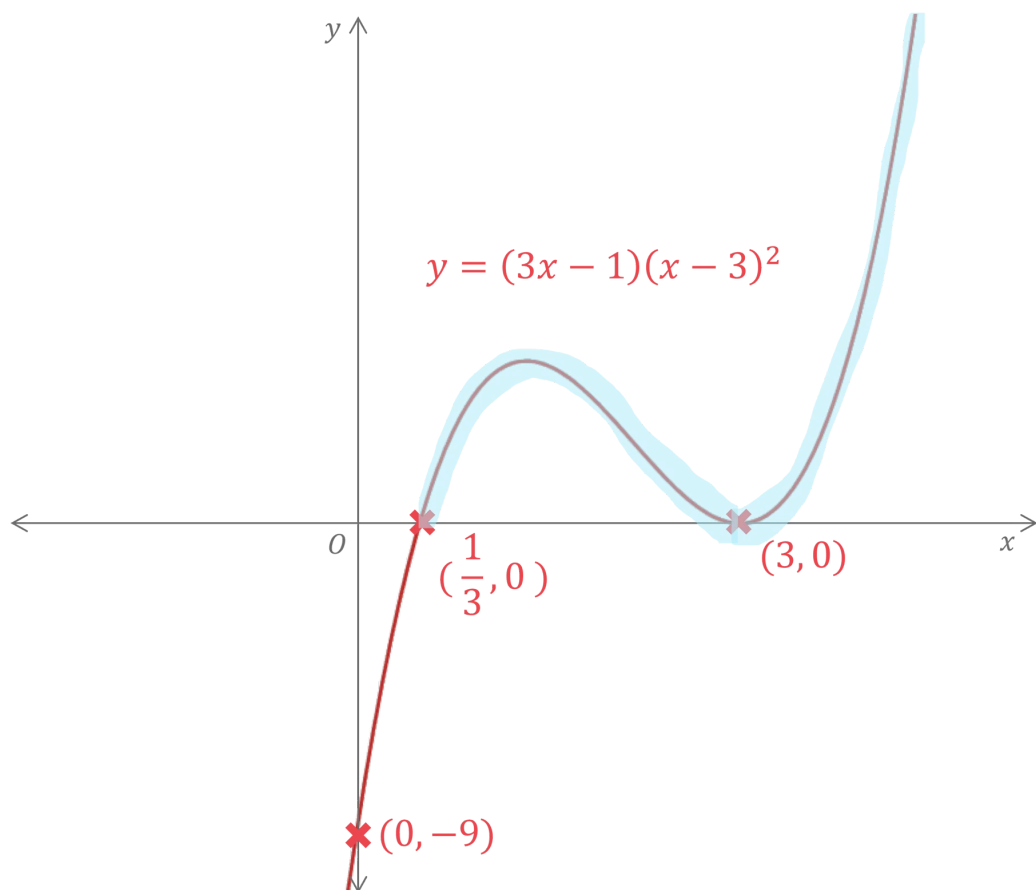
STEP 1 - the cubic polynomial is already in factorised form

STEP 2 - the graph has been sketched in part (a)

STEP 3 - Highlight the graph to show the parts of the graph which lie above zero



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STEP 4 - Write the solution mathematically. Remember that in this case it is a strict inequality, so it cannot be equal to 3 or $\frac{1}{3}$

$$x > \frac{1}{3}, x \neq 3$$